

# Michael Levin & Franz Kuchling ~ Active Inference Insights 015 ~ Morphogenesis

*Insights | Active Inference Insights 015 ~ Morphogenesis, Selfhood, Teleonomy | 1:17:42*

hello everyone and welcome back to active inference insights I'm your host darus P way and today I feel extremely lucky to be able to interview two exceptional researchers Dr Michael Levan is a distinguished professor of biology at T University serving as director of the Allen Discovery Center at tus and the tus Center for regenerative and developmental biology his lab studies anatomical and behavioral decision-making at multiple scales of biological artificial and hybrid systems fr cooking is Mike's postto at toughs with a special interest in the physics of life and how cells adapt to changes in their physical environment guys welcome to the show thank you so much for joining me it's an absolute pleasure and an honor um you're kind of you know everyone in active inference has a foray into theoretical biology but I would say you guys are the first out and out biologist that we've had on the show so I'm super excited to learn um but you may have to excuse some of my more Layman ideas that come from the British schooling system where I wanted to start is probably just putting in some groundwork for the audience uh who might be unaware of certain terms that get used in in the papers that you guys write I wanted to start with morphogenesis what is morphogenesis and in what kind of you

talk about moros space what is moros  
space and how do those two things relate  
yeah I don't know who want to take it I  
guess I I'll start and and then France  
can give his view um so so thanks for  
having us um it's a pleasure to to be  
here and to talk to you um uh there  
there's a there's a fundamental fact of  
biology which is that uh complex shape  
and and functional structure are  
self-assembled so uh let's just think  
about embryonic development we all Start  
Life as a single cell and eventually  
that single cell has to give rise to a a  
a complex organism of different types  
and so this not only during development  
but also during regeneration so for  
example a salamander loses a limb those  
cells have to not only uh restore the  
entire structure but then also know to  
stop when it's done okay and uh there  
are other examples of this of course  
metamorphosis when caterpillar gets to a  
butterfly or or something like that so  
there there are numerous examples where  
the  
biological Hardware is uh is is  
basically reconstructing itself and so  
this is broadly conceived this is the  
notion of morphogenesis now one way to  
look at this is as a sort of open loop  
process meaning that there are a set of  
rules that are executed and parallel so  
um lots of molecule cells and so on are  
are following these rules and then uh  
there's a process of emergence and  
something complex comes out the other  
end you know this is the standard story

of cellular automata or other ways to think about emergent complexity and that's true I mean that does happen and that's certainly part of the story but the more interesting part of the story is that morphogenesis whether developmental regenerative or in the context of cancer suppression or remodeling any of those kinds of um examples one of the amazing things that it can do is get to its target morphology despite various perturbations so if you want to dig in I'll give you I can give you all kinds of interesting examples where not only the environment changes not only injury but actually the components themselves the amount of DNA the type of DNA the number of cells the size of cells all of these things can change and certain morphogenetic systems are amazing at getting where they're going despite all of these things in fact they will often take novel paths to do it which is uh William James' definition of intelligence right same goal by different means so um one way to so so then so then the question is what what conceptual tools can we bring to bear to understand this process uh and what we have settled on or or at least what what we've been we've been uh pursuing for the last some some years is this idea of navigation that what's actually happening here is that you have a collective intelligence it's basically a set of processes um hierarchically organized from molecules to cells to tissues and so on and what they're all

doing is navigating the space of all possible outcomes so there's a there's a huge um you know space of of diverse morphologies that they can achieve and what they're doing is like any agent navigating its environment they are they are trying to get to where they're going and sometimes they have interesting competencies to do it even when things go wrong so now we can bring to bear all of the tools of behavior science of of control theory all this you know cybernetics all the stuff that's used to design um autonomous vehicles you know all of these kinds of things to ask how does the system know where it is how does it remember where it's going and what competencies does it have to get there so that's that's that's why we use the notion of morphospace because it allows us to make testable specific hypotheses about the policies that that are involved in this process wonderful that's a that's a really nice background from which we can kind of explore these ideas yeah it seems when you put it that way very consonant with the the ideas of Act of inference which is obviously something that you guys have have written about before so friend maybe I'll come to you now you there's this idea in these so so people think of priors in sort of statistical space or more broadly about the organism as a whole you know what are the priors of the organism to be welfare to eat and so on but we could also talk about a cell's prior or an a

cell's posterior and basian unfolding  
basian processes within the cell what in  
this idea of navigation and self  
organization what would you say the  
cells prior is is it purely to reach a  
kind of Target morphology or is there  
anything more than  
that I would say there's a lot more than  
that but that's the most um the simplest  
to kind of formulate because morphos  
space like Mike was um explaining is  
something that we we have much more  
access to as a development of biologist  
right so the the challenge is you can  
formulate all kinds of Prior and belief  
expectations for your but if you have no  
way to to estimating how that's even  
being achieved and that's hard to really  
um put in any theory that that's you  
know experimentally accessible but for  
one thing kind to point out that the  
idea of a morphospace or cells  
navigating uh spaces is is only  
challenging for for people that not in  
involved in this field because we so  
used in spatial Dimension as that being  
the space right but you can assign all  
kinds of Dimensions to to a cell we do  
this for humans all the time you know  
the idea of of characters in in a plot  
being multidimensional is something that  
is very much applicable to celles as  
well so to ask to answer your question  
more recently so what what happens in  
terms of expectations of a cell it might  
have you have to really go into what is  
it constantly achieving has any  
relevance for future States so that's La

is can be transcriptional States so what kind of genes are being expressed at any point in time and that is being informed not just by the by the structure of the DNA so by millions of years of evolution it's also been formed by past um events not just in in that individual cell but in it parent cells and and and going on so that's what we call the epigenetic landscape so which genes are actually even accessible for the transcription machinery and which genes are being activated to really transcribe and make certain proteins down the line and so that really is something where where a lot of the pr expectations are going into um that we at least now in the lab are really investigating of what kind of transcription profiles do we see but then also in terms of this it's membrane potential you know and lab studies bi potentials a lot that's something that of course for Neuroscience much more familiar we talk a lot about Action potentials we don't have those in in non- neuronal cells but every cell has a has a m potential electric potential and those are informative and they are um causing lot of Downstream events so we can get more downt online but that's part of what I think is um part the PRI belief and expectations that sell is constantly having excellent yeah um I think there's a really important point there which is kind of something that I noticed reading your papers which is that morphogenesis the the the very idea of physical space seems to be

prioritized um it's self-organization in space is that the only way just in terms of conceptualization is that the only way that we could think of a cell you know reaching a posterior that aligns of its prior does it have to be in accordance with some kind of self-organization space or is there another way that a cell could kind of reach a homeostatic set point I'm absolutely it doesn't have to be just space right so um even if you have a cell just somewhere in the body already differentiated already kind of has it morphology achieved it has to maintain that morphology first of all but then also it has to perform a certain function right your kidneys cells your liver cells they have very different functions even though they're all fixed in space and physical space so PRI beliefs are you know in this case for example for kidney cell is what kind of um what kind of um changes in the blood are being am I being exposed to Parts in the liver same thing um if you have cells in in the brain why it's different they have expectations about Action potentials so um it that space right and again morphos space isn't just um spatial Dimension right it's also it's also traversing because every cell body this an important point every cell in your body has exact same DNA more or less right but what's actually being expressed that's the the difference between all of them so they are vastly different in the actual proteins but

also in the genes are being expressed constantly so when we say traversing and navigating the morphos space we really talk a lot not just about that physical space but how are they changing which genes are being expressed between being turn on and off which really makes the cell identity in the end excellent yeah yeah that that's that's absolutely really really useful to integrate function and space there and I guess those two they feel to me inextricable but actually perhaps so I'm wrong there Mike is it how so so reading kind of the active influence account of morphogenesis we talk about the external states of the mark of blanket for a cell being other cells and in the case of Developmental processes you it turns they end up being other specific type of cells that they then join together with in morphos space is that how how necessary is the presence of specific other cells in the basian unfold for a particular cell or could you imagine that a cell's prior might be confirmed you know absent a specific collection of external cells yeah I mean this is this is a great question and it's it's part of a very emerging research program because it really isn't super clear uh yet uh what precisely cells are paying attention to and and what their internal models of the outside world are what we do know is that in for examp example let's just take let's just take early development so you have a blastoderm and



there are I don't know 50,000 cells  
there and at some point we as as  
external observers will'll look at that  
and we'll say well there's an embryo  
there's one embryo well what is there  
one of right and and if you if you want  
to think about an embryo the thing  
you're really counting when you say one  
is you're counting the fact that all of  
the cells have bought into the same  
picture of the outside world and what  
their goal is so so they are all going  
to cooperate to build a particular thing  
what binds them together into being one  
larger scale emergent individual is  
commitment to a particular model of of  
what where where they um where they  
belong where the boundary is between  
them and and some other embryo opar  
because you can you can make by putting  
in little little scratches into that  
blastoderm you can make twins triplets  
you know there are multiple individuals  
that can come from that Blaster so every  
cell um has to be part of some in in  
that scenario has to be part of some  
Network and that network has to make  
decisions about where it ends and where  
the outside world begins because that  
has to do with size control and and you  
know Left Right asymmetry and all kinds  
of things so uh it's really critical to  
understand what exactly um the the cells  
are paying attention to what exactly the  
uh tissue is paying attention to so the  
cell network because I actually think  
that there are uh there are agents here  
at every level so even actually even to

take a step back you know um even molecular networks within cells I don't even think you have to get to the to the level of a cell before you can start talking about um beliefs and and expectations and so on because as we see even Gene regulatory networks can learn from experience they they can do something like five or six different kinds of learning including pavlovian conditioning and so you can you can ask that question about what does the world look like from the perspective of a gene regulatory Network before it's been trained or after it's been trained and so on right so this is all of course very very active uh areas of Investigation still so we don't know all the answers but but I think I think it's very important to start taking the perspective of the things that we are modeling and asking what do they see and what what are they measure ing what are they paying attention to how do they remap information into Salient um compressions for them you know and and and how that affects decision- making and so on how does that then um integrate with a kind of idea of Telos or tonomy um because you know the way you if we take the perspective of a cell or a collection of cells that are parametrizing beliefs over external States it sounds purposive it sounds like as you say there's agency there um and even the word beliefs people associate with propositional philosophical

beliefs how far should we take that metaphor  
literally well okay so so the first thing I would say about that is just as kind of a ground ground rules I I don't believe in the dichotomy of literally versus metaphorically okay I think all we have in science is metaphor and and and what what you know when people talk about molecular Pathways I mean talk about a metaphor right the pathway relative to what's actually going I mean this is completely metaphorical so all of all of these things are metaphors the real the real question of course is what set of tools does that metaphor give you to make progress what does it help you discover what does it facilitate so uh to me all of this is gets settled empirically to the to the uh to the point that uh if if you have a particular metaphor so maybe you talk about cells as having beliefs or maybe you don't and the real question is what does that worldview allow you to do and I think that's the final judge of it I don't I don't really put a lot of weight into sort of philosophical pre-commitments about what what goals and beliefs have to be it's a question of to Port portability of tools so um I now now having said that so so I think that uh these kind of um uh these kind of tools are extremely useful and they're again not binary because even even as early as the 40s already uh uh weer Rosen Bluth and Bigalow gave us a um kind of a scale of different types of

different goals of different complexity  
right so it's so very simple kinds of  
passive matter and then active matter  
and then and homeostat various kinds of  
homeostats and second order  
metacognition and so you can you can  
sort of go up and up and and it's an  
empirical question which of these are  
the most useful for any of the contexts  
that you study but I think it is uh it  
is absolutely reasonable to see all of  
these things as having some position on  
that scale and you know it's important  
to say that back back before I don't  
know let's let's say before the the the  
the 30s and 40s  
there was a the the problem the problem  
with theology was that uh we didn't have  
a way of thinking about machines with  
goals we sort of Mo most people would  
assume that that humans have goals but  
other than that we really had no idea  
how to cash that out but but it's it's  
really critical to to say that we're  
past that now we we have a good rigorous  
science of uh what it means for for  
mechanisms to have goals it's fine it's  
not magical anymore we don't have to be  
be be afraid of it and um and so uh I  
guess guess the last thing I want to say  
is the difference between thology and  
tonomy so um the def one definition of  
tonomy is apparent gold directedness  
okay a parent thology and a lot of times  
people use that as a kind of um as a  
hedge as a way of softening the concept  
they say well look it's not real thology  
it just looks like Tey you know and and

that's a way so so we don't have to  
really um uh commit to be to being teist  
we're just going to say just looks that  
way so so I like the word tonomy but for  
the opposite reason I am not trying to  
soften the concept I'm I I I'm you know  
not in in in no way uh um kind of  
hesitant about using tileology I think  
it absolutely Works what I think is uh  
useful about the concept of tonomy is  
that by putting the word apparent in  
there what it does is it focuses on an  
observer's point of view apparent to  
whom and I think that is a really  
critical question to ask in almost all  
of the stuff that that we talk about  
here which is not some some sort of  
objective uh uh Universal uh truth about  
these things but from the perspective of  
some particular Observer now that might  
be a human scientist that might be a  
parasite that might be a subcomponent of  
the system that might be some sort of  
superet of the system that might be the  
system itself right we all you know  
systems observe themselves too and so so  
so what that says is that there is there  
is some perspective from which the  
system acts as a goal directed agent and  
thus that could and then can be can be  
used with tools that belong to that so  
um that's that's what you know that's  
that's what I think is the the most  
useful term of uh the sense of tonomy  
yeah France I was wondering what you  
thought about this because um I think  
it's a natural inclination maybe it's a  
romantic fiction that we want to ascribe

some teos to smaller let's say or or  
less complex entities like cells or  
processes within cells there's an  
interesting thing that Mike said that I  
wanted to pick up on which is that  
basically everything in science is a  
metaphor I guess where I might slightly  
push P back on that is the science of  
lived experience um and phenomenology  
which we can't help we have to take in  
some ways literally because it's the one  
thing that we're intimately aware of and  
there's a lovely paper by Anthony Jack  
and Andreas of ages ago 2002 I think  
where they argue that basically you can  
take sort of every psychological  
construct and say well every the way  
that we you know coales on what  
attention means or memory means really  
is because of what we how we experience  
it  
so I was wondering whether there's a  
kind of uh tie there I mean or  
association between when we talk about a  
cell's beliefs a cell's attention a  
cell's memory is there any way we can  
think of that without bringing in our  
own phenomenal  
experience it's tough I mean um like  
especially initially when I earlier in  
my my research when I you know wasn't as  
deep into a lot of the the the  
metaphysics behind it and all that I try  
to really avoid it and I try to really  
focus the my audience on on just the  
mechanical aspects mechanistic aspects  
you know the cell aspects but I found  
that it was almost impossible to do that

because all the words we use are so loaded with everything we have from psychology um but that being said nowadays um I I think really the important part is to do to to do that transversion transversing those two um fields and both ways because the the the claim we're making I think is not that like a seller is doing the same thing on the same level on the same scale like you know Mike really mentioned right there's there's a certain gradient there as a as a human being is but we're trying to understand where the origin of all these phenomena are coming from and are there and and more importantly are the fundamental principles shared between itself and uh and into more classical active INF agent like a human being so things like beliefs right so when you say that S as a belief people like oh well you talking about you know um having religious beliefs and then you get really fastly trapped into into these loopholes but I nowadays try to counter this argument by saying well what is a belief and like where does it come from it can't just it has to a human being has to come from certain expectations right so you really have to then go down to how how do you once he has to Define you have to then go to some mathematical Frameworks eventually otherwise it's tough to do without it and so active inference is nice because there you have very precise definitions of what a belief actually is yeah and so once the kind of trick you can do is

keep a w okay let's talk about Act of  
inference human being need you make a  
certain process you know you have a  
general model General process you define  
the mark of blanket so separation  
between what's outside that you can't  
see and then yourself so an internal  
State and then you have a separation of  
those two by the active and sensory  
inputs you can get right you have  
sensory inputs and then you can act upon  
the environment to change what those  
sensory inputs your eyes your ears are  
are perceiving over time that's actually  
what agency really is about it's about  
controlling you can't really control the  
environment if you can't perceive it  
what you're actually controlling is how  
you perceive the environment but of  
course you're doing that by acting on  
the environment that's what agency is  
sure now once you put it in such a  
fastic framework then the I think  
transferring that to sales becomes much  
more much more easier and much less um  
controversial I think yeah I I I sense  
that that sort of um less  
philosophically Laden more deflationary  
account of agency is is definitely  
possible it's just control in line with  
prior expectations or or you know self  
organiz principles of self-organization  
I think one thing that perhaps Act of  
inference also hasn't got very clear on  
but certainly isn't clear in Psychology  
is selfhood and in some ways I feel like  
it's very fundamental to this question  
of morphogenesis and self-organization



in biological space because it's morphogenesis for who right so likee you sort of noted that in many way well we're talking about that we have it's all Observer relative but also in some ways it's also internal relative because they are man you know they are self organizing for the sake of their own Integrity so well again that might be ascribing a Telos that's unwarranted but it it could be described like that um what is biological selfhood and I guess for people who think of the self as one monolithic thing how does it how does it be how can it be distinguished from more complex forms of selfhood like metacognitive selfhood or epistemic selfhood and so on um I think that uh you know the issue of morphogenesis for whom is actually really important uh because you know a a really defective frog embryo might be a very nice zebra fish embryo and so this whole right and so and so this whole notion of You Know M mutation and uh and just just in general evolutionary change and what what exactly is a birth defect is really important and interesting chemistry doesn't make mistakes so there's no such thing as as a mistake in chemistry but there is potentially a mistake in carrying out developmental biology depending on what perspective you're looking at so it's something very interesting is you know the the ability to make mistakes is a Hallmark of a cognitive system right because because that suggests that you had certain

expectations that were not met and so on  
so just the very nature of a of a birth  
defect is is interesting so now now the  
notion of a self uh I of course there  
are many ways to think about it I I'll  
give you what how how I've been thinking  
about it um what one of the things that  
that I really wanted to do uh early on  
is to come up with a framework uh to  
directly think and compare think about  
and compare agents of very diverse  
structure and Providence so we're  
talking about you know conventional  
beings like like us and animals I mean  
also those are also of course Collective  
intelligences right we're all made of  
cells and so on but also you know new  
synthetic things that are going to be  
created um robots AI hybs h ibds cyborgs  
you know all kinds of mixed kinds of  
things potential exobiological life and  
so on what what do they all have in  
common you know what what do all agents  
have in common no matter how they're  
made and no matter how they got here you  
know what their origin story is so what  
what I settled on and and so now you see  
you know I'm super into the whole  
thology thing so so what I settled on is  
this notion of um the size of their  
goals in other words uh I I Define this  
notion of a cognitive likee which is um  
it's not the dis that you can sense or  
act it's the size of the goals that  
you're able to maintain the biggest goal  
that you could possibly maintain is the  
size of your cognitive likeon so you  
know so so every every agent has some

degree of memory going backwards maybe  
large maybe small some predictive power  
forward and some spatial area of concern  
over which its goals play out and um and  
so so now now the thing about that is  
that that is the the determination of a  
cognitive lyone by an observer for  
example here you as a scientists are  
given a new system you don't really know  
how it works and you want to do some  
behavioral experiments and you find out  
that here are the things that it can  
possibly care about that that is a a  
third-person investigation of its agency  
and we say this is an agent to what what  
I think of as a useful definition of the  
self is exactly that process in first  
person so a self is what how that  
process plays out for the system itself  
so the system itself also has to be able  
to corar grain the outside world and not  
only tell agential stories about  
um things in its environment that do  
things right because no no system can  
afford to to track micro States you know  
lassus demon isn't going to you you'll  
be eaten and and dead long before you  
calculate anything so so every every  
agent in the real world has to has to  
course grain and they they tell these  
stories about you know agents other  
agents doing things but you also have to  
tell that story about yourself to some  
extent you you can't avoid it because if  
you don't have a compact uh self model  
you are not going to be very good at  
controlling your own Parts in making  
things happen it's just right that's

just part of being effective in the real world and so I think what we say when we um what we mean when we say self is the internal first-person view of a system about its own right so so so so you so so there's kind of a bundle of things you need to have you need to you need to have a sense of boundary between you and the outside world so which are the things that I think are actually me versus versus not me you need to have some representation of some space that you think you're operating in you need to have some sense of your own sensors and effectors what what can I actually do what am I actually sensing and so on we as third person Observer scientists uh try to tell those stories about other things that we work with but but agents tell them about themselves and so do par you know parasites and comp specifics and various you know potential mates and prey and cheaters and and all these things everybody's hacking everybody else and in order to hack other other agents to hack yourself and your own parts to have control over your own Parts requires that sense of selfhood and it might be you know as as France said the point isn't to say that every Rock has human level hopes and dreams that isn't the point the point is that uh there there's a huge range in how much processing is how much sort of meta processing is done about all of this you know you can you can roll all this forward in a in a very kind of minimal way or you can have all kinds of Loops

where you're actually monitoring your own progress on certain goals and you have the ability to change your goals and to pick new goals I mean there's a you know you can layer stuff on and on but that's yeah that's really great that's a really lovely explanation yeah it really deeply aligns with the way that I think about selford which is fundamentally from a sort of mearian um phenomenal self- modeling aspect and that's exactly what Thomas would say which is that self-modeling is the capacity to recapitulate or res simulate the actual data structures of your own body in a kind of Co grain space so it's you know the the one thing that came to mind is talking about the uh cognitive light cone is in terms of the the extent of your goal yes it's in space but it's also in time and that feels like a very relevant thing here um when I've spoken to Carl and and when I've seen the direction of active inference it's very much going in time uh in in you know the direction of temporal planning deep temporal planning which arguably relates to the depth of your generative hierarchy because the deeper your generative hierarchy you know the the highest levels are going to be tracking the most invariant and long-term fluctuations in the external environment so friend I'm just curious if you're I mean this is just a thought experiment is there a way where perhaps if you have a a cyborg or a tad Pole or you know some entity that you're examining and

you're trying to get to its cognitive infrastructure is looking I don't know whether this would be behaviorally or anatomically is looking at its capacity for planning which might well just be its the depth of its generative hierarchy does that give us an insight into its actual

you know existential or or experiential life I think it does and it relates back to the your early question about what are its actual beliefs you know what on what level are its beliefs being expressed and so I just gave you a couple examples what I left out is you know that of course like you just said what's important is the hierarchy of that how are these different different beliefs organized in a certain hierarchy of spatial and temporal scale and also to to I really like the point Mike earlier by chemistry not making mistakes right so what that means is the conclusion of that idea is that any purely chemical system is not going to change much it's become entirely predictable it's not going to push back on anything in the environment it's going to have a certain it's going to come to an equilibrium pretty pretty fast usually and then that's that's it and so another way to form this question of how does a cell how can we um test agency to answer your question how how we test um having a self certain certain ability to form hierarchy and and form beliefs we have to understand of how much push back can it can it give you

can it give to the environment over over  
long times and so the way I would probe  
this hypothetical cyborg or or tadpole  
in terms of like what kind of um  
expectations what kind of planning is it  
is it doing is you have to really see  
how these different parts of that  
generative hierarchical model are  
interacting so you will have a you have  
a quick first response which could be  
you know purely physical but then very  
often on turns very active in most  
system that's the the active inference  
part is really what makes life life take  
you know it's it is entirely active and  
it's active very quickly but also over  
long time scales and so um what we're  
trying to show in this one paper by  
recognition with Chris fields and Mike  
was that that's a drive that got  
selected for very early on in in  
evolution and so the way to Pro the  
system is really looking having  
different reporters um generically but  
also just having certain dies in A  
system that will give you readouts of  
its its active States like what is it  
making what kind Poli is making is it  
changing it's I of potential and then  
you try to relate those things are they  
completely independent are they just you  
know a non non living system you expect  
those to be pretty pretty static with  
certain very predictable Dynamic um  
reoccurrences of patterns but in the  
living systems they're going to change  
and they're going to change dependently  
on the scale so one scale is going to

affect the other one and it's going to continue to doing that and that's I think how the planning is achieved by that temporal in uh independent inter interdependence between those different aspects of a cell if they were independent they would be no planning but if they're dependent of on each other then the cell is basic going to be using its long-term processes to redirect and inform it its active points at any moment in time nice yeah yeah yeah and speaks to a kind of almost predictive coding hierarchy um where you have these kind of trickle down predictions Mike did you want to add yeah just I just want to go back back for a second to something you said before which I think is quite interesting um about the uh the you know the the the non-metaphorical nature of our phenomenal experience and I I I I take your point and I think phenomenal experience is really important but but there's there's the issue that we change right and so and so how you inter so so I mean for us so let's say puberty is a is a is an example right where where a lot of veence changes and things that you thought were really important suddenly not important at all but all this other stuff is now important and the the example that I really find uh kind of in meaningful for all this is the caterpillar to butterfly transition so what happens there is you've got this you've got this softbed creature that has a that lives in a two-dimensional



world and it crawls around and it has a particular controller for that kind of soft body you know actuation so it's got a certain kind of brain and then it has to be transformed into a completely different creature that's a hard-bodied kind of vehicle it flies it lives in a three-dimensional world and so on and so so it has a completely different brain and during that that process the brain is basically mostly broken down most of the cells are killed off um and then reassembled you know it grows it grows a new brain so so there's there's sort of three levels of interesting things here the the first level is that that that aspect of phenomenal experience like what's it never mind what's it like to be a butterfly but what's it like to be a caterpillar becoming a butterfly and how much of right and how much of the um the metaphors that how much of the things that the caterpillar took very seriously the butterfly thinks were total metaphors at this point right if it could I mean I'm not saying that's how it thinks about these things but but right things that were quite um quite concrete for the for the for the caterpillar are no longer so for the butterfly that's the first thing the second thing is that um we know there's continuity you know if you if you like um the stability of memories as a Criterion for selfhood which a lot of philosophers like uh we know that the butterfly remembers things that the caterpillar learned so there is some

continuity here even though the the  
brain cognitive system in the body of  
all of that stuff is radically changed  
but there's some continuity and so the  
next kind of question you think about is  
okay where the heck is this memory where  
when the that survives total brain re  
you know refactoring but the third the  
third thing that I find even even more  
important about all of this is is comes  
from the simple fact that um the way the  
way the way they test these memories is  
well one one way to do it is you um you  
you feed the caterpillar leaves on a  
particular color disc and then what you  
find is that the so it forms the  
Association and then you find that the  
butterfly goes back to that disc to feed  
right but the but the key the key thing  
here is that butterflies and  
caterpillars don't eat the same stuff  
right the caterpillar eats leaves the  
butterfly doesn't care about leaves the  
butterfly wants nectar so um so now  
something very important you cannot just  
you know Persistence of of of the self  
in terms of memories what you can't do  
is just keep the memories constant  
because you'll lose the salience they  
make no sense to those memories will  
make no sense to the butterfly and in  
fact what you have to do something very  
interesting you can't learn leaves on  
this color you have to generalize to the  
to the to the higher category of food  
and so right and and and then you have  
to remap so this is the part that I'm  
super interested about in terms of um

moving memories around and and and  
transplants of memories and you know  
personal identity and all that you have  
to take the the Deep lessons of your  
past life and sort of re uh reinterpret  
them for your new life which is by the  
way completely different anatomy and all  
of that but there are some things you  
learn that are still pretty useful just  
not exactly in the way right it's not  
the it's not the details that that  
matter now it's the kind of the generic  
concept that matters so so re remapping  
right remapping that information onto a  
new exper onto a a a new um embodiment  
for for for that for that being that's  
that's why I think the concept of  
metaphors is still super strong even for  
the phenomenal stuff because because you  
can't take any of it literally because  
you're not going to be the same later I  
mean you know as certainly for some  
animals it's more more than so than  
others but you you have to understand  
the Deep metaphor of what you learn not  
the not the specific details of your  
experience yes yes that's really good  
that's really nice there's a lot there  
um yeah it again it's converging with  
some of the stuff that I've been  
thinking about and writing a little bit  
in draft which is I'm kind of I'm coming  
to this idea that the self in some ways  
is a is a relevance filter a salience  
filter such that CU I think at a very  
deep phenomenal level everything that  
Finds Its way into attention or  
awareness is relevant it wouldn't really

make sense and so I I would say that the  
one non-metaphorical thing there  
although you have the higher order  
concept of food is  
relevance and relev and meaning making  
to me and maybe well I I it's  
interesting you say you know we're  
trying to go up in abstraction up the  
generative hierarchy the one thing that  
the butterfly and the Caterpillar share  
is the imperative for  
nutrition and and maybe that's maybe so  
when we can start linking things in some  
kind of map of similarity maybe the  
thing that unifies them is  
preferences um so the caterpillar and  
that maybe links us with the caterpillar  
and the Butterfly right we have this  
unified um goal that said the thing you  
said about the fact that they return to  
the same kind of leaf is very  
interesting because there does seem to  
be a idiosyncratic maintenance of memory  
uh which I can't quite account for there  
it made me think of Andy Clark's paper  
um knitting your own Markov blanket  
where he talks about the um  
metamorphosis of Markov blankets and how  
we can kind of get around this problem  
of like when is one thing another thing  
by adopting a process ontology rather  
than a substance ontology but friend or  
Mike you know whoever's got an idea on  
this is it is it possible it's just a  
false move to say that the butterfly and  
the caterpillar are the same thing like  
that's the same organism we not just say  
that you know there actually does become

this division or this separation at and we it's philosophically vague as to when it happens but there are you know there was one thing and then now is another so so so I think the way uh the way to avoid these kind of philosophical traps is by being very practical and asking why why do you need to know so so if the right so so if the reason you need to know is is because you want to predict how much crawling around there's going to be then it's absolutely two different things because there's going to be zero crawling around afterwards but if but if what you need but if what you're trying to predict is um so what color substrate will it like to feed on then then then then you need to understand that it's the same thing because you will have less predictive power if you don't understand that it keeps memories so it's just again I think we're back to again the the the um the perspective of of uh you know what what question are you asking as the Observer yeah okay cool let's let's park that there I think something that you've worked on Mike but also France you mentioned and I want to pick up on is the difference between neural and non-neural structures um there's a there's a kind of neural bias in Psychology Neuroscience a quco bias um France one why is that the case do you think and two why should it not be the case if it should be the case maybe it should be the case yeah I mean one uh human

Centric because we are humans but more  
more fundamentally the reason that that  
that we we are so confined on the idea  
of of neuros circuits being the only one  
that doing any inferences because  
they're just very fast right Action  
potentials are fast it's a highly  
optimal optimized  
Network optimized from what right  
neurons didn't come from nothing Asos  
didn't come from nothing electric  
potentials were were being used by cells  
way before the U innovation of of  
neurons so that is I think the reason  
why now we're trying to look at what by  
a neural organisms and just cells of of  
having these same properties um and more  
and I think other than just that being  
kind of transition what's also really  
interesting is Mike and I both it's one  
of my favorite um past daytime  
activities is asking neuroscientist what  
what about your theory is specific to to  
neurons and it's a question that nether  
of us has have have gotten a good answer  
to yet because a lot of it of course you  
know there are certain certain time  
periods it takes if you're looking at  
fmri Imaging looking at the the  
oxygenated blood flow in the brain so  
you have to know those those time  
constant how fast does it take if you're  
looking at if you're actually measuring  
act neuron you need to know all these  
transition times to really make your  
model take but if you change those time  
constants and you compensated for that  
in your model would still work so as far

as we know um so far we're happy to get proven wrong on here but so as far as we know there's nothing that specific about neurons to that you need to make any of those models including influence actually work they highly optimized and they have completely different behaviors of course um because those timings do matter but they're not necessary for it because they're not NE they're not influence framework right it's not necessary to have a brain to have a model blanket it's not necessary to have a brain to have a generative model so once all the actual assumptions that go to a model can be applied to any real systems then you're good to go and you should apply it to those systems as well yeah I guess another biased mapping here is between cognition and the Brain um and there's I feel like there's been a a you know a movement towards obviously in terms of embodied cognition and the kind of dynamic the agent Arena Dynamic that was happening but also more recently might your paper with Anna Ana Chona uh the brain is not mental you sort of I think you look at the immune system and sort of say how the immune system contributes as well as the neural system in subserving um adaptive control self-organization perhaps you could just speak to that and and say you know well I mean it's a it's a guess it's a fault experiment what would it look like if we didn't have the immune system if we were just trying to self organize purely in terms of

neurodynamics yeah um I I mean we certainly there are uh there are knockout mice where the immune system is taken out and all that kind of stuff so and and and a lot of self-organization does happen before the immune system kicks in I think you know I I think the idea is simply that uh there are many many different kinds of systems that go into this Collective that that we see when we say it's it's it's a human or or whatever and and all of them are potentially important contributors to the to the process um as as France said you know uh it's it's it's a really interesting experience to uh to ask people what a neuron is and and and and watch them list things that are basically every cell in the body does this and and then and then you know so so I have another thought experiment that I sometimes do is like like back in the um you know back in the day when uh when when the idea of neural networks was first being worked out right um I wonder if if we had come along to uh to some of the some of the greats uh in that in that field um and we said uh oh by the way just just FYI um the biologist got it wrong thinking is in the liver not the brain so so there's two there's two ways they could have gone right one way is they could have said I don't care what you say we've got model of cognition and they require it to be to be neural so you guys must be wrong or they could say fine what do we care are are this what you know the the



the basic calculus of you know mullock  
pit all that stuff the basic calculus of  
it doesn't require doesn't care what it  
is so fine it'll beiver right so so I I  
tend to think it would be I tend to  
think it would be the latter you know I  
don't see really anything in the in the  
Deep In The Deep lessons of um  
behavioral Neuroscience that's really  
about neurons per se you know it's about  
multiscale control it's about active  
inference it's about all these kind of  
deep things that go far beyond brains  
and they're applicable to all sorts of  
stuff stuff well are there certain  
processes which are reserved for the  
brain I mean I think someone listening  
to this probably a psychologist or a  
neuroscientist will say well take the  
brain out good luck take the liver out  
you know I mean still good luck but you  
might have a better chance thinking  
right uh for for a short amount of time  
I mean take the take the mitochondria  
out and uh and and you're done for  
pretty much at the same rate as as the  
brain now look the bottom line like I  
nobody's arguing that the brain isn't  
unique and interesting in certain ways  
right like we F you know as much as we  
find uh protoc cognitive capacities  
elsewhere I've never made the claim that  
language or longform planning is like  
we've not found it could be you know I  
wouldn't be terribly shocked if it were  
found but but we haven't found it so we  
don't make the claim so clearly brains  
are interesting and they do they do

interest things but um I think it's important to uh make make a distinction between between having found unique features and told a very specific predictive story that that why of why that is versus just this generic gut feeling that everybody has that that I have real you know real feelings and plans and this thing is a slime mold that I mean I don't know how many times people have said to me well that's a slim mold that's not a real decision it's just physics and and it just it drives me up a wall because what do you I mean you don't think we could tell a physics story about you like of course we could it wouldn't be a very good story but but neither is it about the slime mold and the fact that you can tell that story doesn't mean anything there's always a physics story to be told about anything and so you right I think I think it's it's fine to have specific theories about what it is the brain that's doing differently than other structures that's great um but a lot of the resistance to the kind of Continuum views that that France and I are talking about comes from a kind of closet dualism it's people really you know they just really feel they're different and and and they have to accept the fact that the physics are not different so well now what is it then right it's like this then then there's something else and I think not not even that I'm particularly against that I just think it should be like come you

know dragged out into the light and  
let's just say like say what you think  
the difference is you know I think I  
think that's really critical see I'm not  
sure the difference I'm not sure the  
jism really fundamentally is a uh brain  
body jism I think it's an awareness body  
dualism or Consciousness I mean like a  
traditional cartisian dualism I think is  
really at play here it's just we're  
we've been persuaded that Consciousness  
is rooted in the brain and so once you  
make that Association then you go well  
obviously my brain is special because it  
generates Consciousness Consciousness is  
special I can in principle imagine  
myself being conscious without my liver  
my leg my arm and so on and so that's  
the special part yeah is am I what am I  
missing there no I I think you're right  
but but I think I think I think what  
people are missing there is when they  
say conscious of my liver that isn't  
that isn't the question here the  
question so so right now your left  
hemisphere and my left hemisphere are  
having a delightful discussion about why  
Lang you know why Consciousness is in  
the brain and uh and and how important  
the brain is for for Consciousness  
that's fine then I will say to and I  
don't have a new Theory Of Consciousness  
by the way so I'm not like I usually  
don't talk about Consciousness at all  
but but since we're here I'll I'll just  
say that um I I have said that for the  
same reasons we think that the brain can  
support Consciousness we ought to take

seriously that other parts of bodies  
also support Consciousness now at that  
point people say exactly what what you  
just said which is well I I don't feel  
like my liver is conscious but but  
that's a huge mistake of course you  
don't but you also don't feel that I'm  
conscious right that that that's that's  
that's exactly right so so so it's nice  
that the left hemisphere is eloquent and  
it has language and it can tell stories  
to to other left hemispheres about how  
how you know how conscious it is but  
that doesn't mean that what you don't  
have spread throughout different regions  
of your body is a non-verbal  
non-linguistic uh Consciousness that is  
not for us to access anym than we can  
access each others and that's that's the  
you know the fact that you don't feel it  
is no surprise there's no surprise there  
um you have to take its perspective and  
ask what do you think that's going on in  
your brain and so so literally um a  
student and I have have been uh we have  
this um we have this table that that has  
rows for the main theories of  
Consciousness and for each Theory there  
is something specific you know so so so  
Stu Hammer off will say it's  
microtubules somebody else will say it's  
you know electromagnetic field somebody  
else will say it's you know integrated  
information like what whatever it is and  
then and then we just look and say okay  
given that where does that occur in the  
body and the answer for almost all of  
them is everywhere and so none of these

theories as far as I can tell can distinguish you know uh and then and and some of them try to rescue it so right so so IAT has this postulate that says okay but but but there's there's we're just going to say there's only one right that that's that's that's a postulate you sort of add on to things right that's not so so that's that's the issue is that it's not about you feeling your your liver being conscious it's the first person perspective which may or may not be there yeah there's plenty to that I actually yeah I I'm reticent to dive into Consciousness because all my podcasts end up being about Consciousness um and I have biologists on I don't want you know but yeah it's it's it's interesting I mean obviously a lot of this is being we have to take as a kind of um Axiom that at least for this conversation that Consciousness is uh well generated in the brain whether that's in a property je list or whether it's in a productive way and I also do just want to give a shout out to slightly more esoteric forms of idealism conscious realism and so on but it also makes me think about let's if we just get away from Consciousness in terms of first person perspective and just talk about basil intelligence in in in terms of morphogenesis it also makes me think well it's this is a term for Consciousness but people will talk about being whether conscious can be is multiply realizable has multiple

realizability I.E if I put you know you  
know CH you know if I had a Chinese  
Nation their blocks idea with exactly  
the same functions as my brain exactly  
the same synapses because there's the  
same you know those connections would be  
the same between the members of the  
Chinese populace and my brain would I  
have Consciousness okay if we get away  
from consciousness

I guess my question is is morphogenesis  
itself can it be is it multiply  
realizable in the sense that we're now  
talking less about the human body is  
there something sort of or and more  
towards like cyborgs or robots what you  
know people have this intuition I think  
as well that there is a life force and a  
I vital that leads maybe to this  
self-organization right there like  
there's got to be something there like  
or else why why are these things  
resisting entropy why are they  
self-organizing and I think the idea is  
like well if I just stuck some St of  
Prior in a robot why on Earth would that  
end up being self like why would that  
work to use very reductive language  
friend I'm just curious is there like an  
added ingredient at least it might be a  
metaphor but an added ingredient that we  
have to put into a non-carbon based  
thing in this case to to give it the  
chance to self  
organize yeah probably I would if I had  
to name be Evol evolvability um that  
would be one one part so if if the and  
it doesn't have to be evolution of of

its actual physical components that can be evolution of its of its programming as well but that would be an important important point to it um the one thing I like to point out in for that question is right I I talked a lot about in this podcast about one my one of my biggest interest is really understanding of where that drive for for an agency comes from but that's not just a kind of a philosophical um interest kind of it's actually really once once you find sufficient evidence for that being a fundamental Drive in evolution then you have to accept the the to combine that fact with the idea that well if any part of your body or any any robot um that you're putting into this is going to strive to maximize its agency with respect to its environment that is the real reason why if it turns out the liver is not conscious um right now then the answer would be if you accept the fact that well the drive is fundamentally in evolution any any life system is going to try to maximize its predictability and its agency over its environment then the answer would be well its environment isn't very interesting so the liver probably didn't have any any reason to maximize that drive if you were to put the liver in or you were to put a robot into an environment that's very active very much changing very volatile and you give it enough either enough time to evolve if you Comm to vulnerability or you just put in um a high degree

autonomy over its own beliefs then over time with that driver it should be able to to to maximize that the problem right now is that the way we program robots is uh is very much right it's it's an very clear algorithm we don't allow any adaptiveness to it um and so that's that's how you would have to go about it I think and you may not want to do that just FYI but because the problem right is um you know we talk about one of my favorite

um uh examples of why what difference between machines and and a brain is that a calculator is infinitely better than you at calculating you know adding numbers together multiplying it I mean you know and it's the dumbest thing ever like it's so simple and we've had these things for for decades now but so if you want to have a predictable reliable answer it's always correct you know the the human is not the best best way to do this that is the reason why we're not even if we had the theoretical framework to make a really agentive um intelligent system um for what we want them to do which just really make our lives Easier by making the same thing every day reliably you wouldn't want to do that but if your goal was to create a new uh intelligent um um cognitive agent then you would have to go about it differently yes that's an interesting little Paradox isn't it I mean like the most robust thing in the world might be a calculator but it's s quite useless it's like the more useful you become the



more uh precarious your existences um  
Mike I was wondering if you had any  
thoughts on that kind of multiple  
realizability question and I guess what  
I'm going at with that  
is it's a very it's a it is a  
philosophical question but I think it's  
interesting for biologists which is why  
would Evol like why would Evolution  
prioritize self-organization I mean  
there just a fundamental question there  
in terms of entropy I mean why do we  
have these pockets of neg entropy  
entities um is there a biological answer  
to that Beyond just well to be a  
biological entity you are neg entropy is  
is there anything can we get anywhere  
beyond that yeah um a couple couple of  
things I think are interesting and I  
want to be careful here because I I did  
I did a little while ago I did um I  
started writing a paper on what exactly  
is missing from our technology to  
provide true agency because I think we  
can actually say what some of those  
things are and it it quickly sort of  
dawned on me that to whatever extent I  
was on to something that would actually  
lead to a massive creation of of new  
beings with that that matter in the  
moral sense and not I'm not too  
interested in in you know being  
responsible for that so some other  
people will do it I'm sure but but but  
but I don't I don't want to do it so so  
so um but but I can but I can say a few  
things um I think I think uh when we I I  
I when when we make the distinction

between life and Robotics um even aside  
from all the hydroids and cyborgs that that  
show us that it's like a Continuum but  
even aside from all that when we make  
that distinction here here's what I hear  
I think life is what we call things that  
are good at scaling up uh goals so so  
you know when you have very when you  
have a bunch of subunits that have very  
basic competencies you know may maybe  
all they can do is follow least action  
laws or like like really simple  
things when when when you've got a pile  
of them and the pile has basically  
exactly the same cognitive lone as the  
components we say well that's  
not living that's a rock that's just you  
know but but but life is what we what we  
call things where there are um special I  
I I've also called it a cognitive glue  
there are like these special policies by  
which the parts relate to each other  
such that the collective has a bigger  
cognitive lone in new spaces that the  
parts couldn't have and that's and  
that's what we call life now now under  
that under that way of thinking about it  
could you make that out of other things  
I'm sure you could and I and you know I  
tend to think that in the wide universe  
I tend to think that there's probably  
lots of extremely diverse examples some  
most of which we wouldn't even recognize  
of of of that happening and I think um  
specifically I think I think what  
happens is that uh the The evolutionary  
process really I I I don't really think  
it makes specific solutions to specific

problems the way that we do when we do genetic algorithms and things like this I think what what biological evolution does is it makes problemsolving machines and that's because if you overtrain on your evolutionary priors and you really take seriously uh what expectations that you might have about what genes you have what the environment is like if you take that too seriously you're done for because because guaranteed and this is why I also think that regeneration is not really about repair of um of physical damage like like external damage I think fundamentally regeneration is about knowing for a fact that your own parts are going to change right EV Evolution just means the fact that you have a so some kind of a lineage that's subject to mutation whatever you you know your stuff is going to change if you don't accept that you're you're never going to survive long term so so what ends up Surviving are architectures that in particular are these multiscale architectures where the individual components have agendas and then this you know they they make up systems that have other agendas and and and on and on our our current robotics by and large is very flat right so you hope your robot is intelligent but the but the parts it's made of are not they don't really tend to do anything on they don't have their own agendas and I've given a talk called that why robots don't get cancer and this is why right it's because you they're made of parts

that are not that that don't have their own little little goals in other spaces and there's no danger of them defecting unlike biology where it happens all the time so um I think what's I think what's happening here is that we end up with these architectures that uh are by virtue of cooperation and competition of parts that that end up being uh being having a CO a larger cognitive lonee you end up with this kind of this kind of intelligence ratchet and we could you know I could I could tell you stories about how that that type of thinking explains you know planarian the the amazing facts of planarian regeneration and so on this this this idea that what evolution I think really selects for is the plasticity to and and and and it works out in different types of animals to different degrees but but overall the plasticity to deal with novelty you know this idea that when you when you emerge into the uh into the world you don't know for a fact what you are what you have is your genome the same as it was are your cells the rights shape or size do you have are you in the same environment you don't know any of these things you you have to um do your best in putting together a model of what that that that's why life is so interoperable that's why we can make anthr robots and xenobots and interface living tissue with crazy nanomaterials and make these Hots that are you just sort of totally cobbled together it's because by and

large it's already bought into you know  
the the life we see today has already  
bought into the fact that um like like  
you know everything changes you know and  
you and you can't you can't depend on  
these things you have to figure them out  
in real

time yeah yeah great it you know when  
you mention the word cognitive glue it  
made me realize that there's one thing  
that we haven't even touched upon at all

I haven't even mentioned which is  
electricity and

bioelectricity um so I think actually  
you had the paper mik didn't you last  
year two years ago um about  
bioelectricity being the glue of the  
cognitive mind soort of from physiology  
all the way up to the mind and obviously  
this is very much you know a seminal  
piece of your work is the processes of  
bi bioelectrical processes in cells and  
outside of cells in terms of self  
organization and I think at least to my  
very Layman biological is electricity  
almost has the the kind of center of aam  
v a kind of thing that you know animates  
things and maybe you know I'm thinking  
about

Frankenstein um either one of you I  
don't mind to everyone

start what you know actually friend it  
was something that you mentioned which  
is maybe one of these ways that we can  
conflate the neural nonneural divid is  
the fact that both of them have both of  
them fundamentally

electrical what what is it about that

electrical ontology that animates or my  
animate intelligence or  
morphogenesis so when I was writing my  
my PhD thesis I was trying to answer the  
introduction the question of where does  
where does neuron even come from and the  
the best paper I found that was  
answering that question was looking into  
this very small organism that had just  
bunch of vdia around its surface it was  
a marine organism that was living on the  
bottom of the ocean and it was trying to  
Center environments and it by small  
changes in electrical fields that that  
were happing around it and that was  
causing um like many I won't say action  
potential but it was causing basically  
many vibrations and because us that to  
change change his movement so the reason  
I think why that would you know why  
electricity is that so important here is  
just very it's very simple mechanism to  
have that is physically a lot more  
accessible to all parts of the body  
that's actually the primary reason why I  
joined Mike laab and why I'm I've become  
a biologist I was a physicist before so  
I was looking for something in biology  
that would be all these would not be  
subject to completely different  
mechanisms if you switch organisms so  
like the moleular pathways we are  
typically investigating as biologists  
they are of as as they were before they  
are now still very important in all  
these different functions of the cell  
and organisms there kind of somewh  
unsatisfactory for the physicist because

it's just so messy right you can't  
really make heads or tails of it and  
there is no clear field of potential  
that is is just one number essentially  
right you want to dice it to something  
very simple the beauty of using bits is  
that's you can you can have one number  
one potential you can like a me  
potential you describe to cell it's  
still made up of all the individual  
Parts there's many iron channels  
involved in this all the molecular  
pathways are still making up they all  
feeding into that one um one number that  
one B potential but what's more  
interesting is the red part right you  
saw if you have to constantly compare a  
millions of different molecules in a  
Cell that's not going a good good  
formulation for for achieving  
Consciousness for even doing any smart  
Behavior having something that you can  
constantly having a surface almost like  
a space that can read they can write  
onto and read out to I think that was  
makes electric potential so appealing  
that the only question then is like why  
wasn't it other physical Fields like  
mechanics they are of course very  
involved as well in developmental  
biology and regeneration as well but  
mechanics don't have this neat feature  
where it's not as as fast and it's not  
as localized so basically the way I see  
it the potentials have a have a  
beautiful tradeoff between being  
physical having a broad readability on  
on a cell or even a tissue level but are

confined enough that they can be modulated enough to actually form active Behavior which you can't say for most for the other physical fields that we would um think about in this context great Mike you got anything you want to add

yeah yeah a couple couple of things one one thing to add is that if you think about um one of the really powerful things about um bioelectricity is that imagine an ion channel is specifically a voltage sensitive ION channel so as soon as Evolution discovers a voltage sensitive ION channel what you have there is an element with historicity it means that uh the current state is a function of what was going on before and it means that what you really have is a voltage gated current conductance AKA a transistor and so so so you know we know once you have that then then you can do anything and so uh electricity is is really good for these kinds of feedback loops right both positive and negative it's good for um representing information across time and space it's good for functionally integrating a system uh across across distance some some of the um really cool early uh evolutionary data on this come from bacterial biofilms and this is um G so's amazing work uh what basically he's he's got a paper called them you know brain-like signaling in bacterial biofilms and the idea is that already in a in a mat of bacteria so like a really long time ago um Evolution



already uh HD onto this uh to this idea  
that um by using Electrical uh  
electrical signaling to integrate  
activity among competent subunits you  
can get gains at the collective level so  
they take turns it's used for for  
nutrient sharing and and and things like  
this but but you get these Collective  
Dynamics and so I think it's just it's  
it's a very convenient modality for it  
that was that was found a long time long  
before neurons you know picked it up um  
it was used for that kind of thing but  
but no doubt there are other there are  
other ways to do it somewhere out in the  
universe there there there's other I'm  
sure I'm sure there are other kinds of  
cognitive glues out there but but here  
electricity works really well cool and  
so I know we need to wrap up relatively  
soon  
um and I guess it's it's a it's a kind  
of dark pun but us wrapping up kind of  
soon makes me think about aging and  
death and Decay and and whatnot and  
there's a there's a massive fuss about  
aging and Longevity at the moment um and  
David Sinclair and this notion of  
epigenetic scratches and it kind of  
makes me think of the inverse of that  
which might be regenerative biology and  
the capacity to reestablish  
self-organization reestablish uh the  
kind of priors that we've been  
discussing in terms of the functional  
priors the the preference priors the  
ones that you know are seeking a certain  
sort of uh external state or seeking a

certain probability distribution so as  
to be able to fulfill its function so to  
speak um  
what Mike what is the kind of lay of the  
land in your world as to  
aging um how seriously should we take  
this kind of epigenetic approach I won't  
ask about the moral aspect of it because  
it's a completely different question  
about whether we should live longer or  
not  
but sure you know having localized uh  
morphogenesis and self-organization is  
really the currency of biological  
systems how viable is it that we can  
reest that in something which is you  
know where self organization is ailing  
and and rather feeble yeah um I think  
well the the first thing is I I do just  
want to say something about the moral  
question because it also gets to the  
issue of regenerative medicine more  
broadly uh and uh and and Beyond  
regenerate medicine human augmentation  
and all of these kinds of things so so  
the one the one the one thing I want to  
uh point out is  
uh if it were the case that we were in  
some way optimized by some intell  
elligence that shared our values then  
one could make the argument hey let's  
not push too far off of that right  
you're you're screwing up the planet and  
I get a lot of these emails from people  
saying uh well well things are great you  
bet you know you scientists better not  
mess it up and I just want to be clear  
that there there is no basis for for

this for this view we our our life our health span our aging our susceptibility to to all kinds of dumb diseases and our lower back pain and our stigmatism all this all this stuff is is not in any way uh o optimized for any of the things we care about that's just where Evolution happens to have dropped us off at this point that's that's it and so there's nothing magical about this lifespan if if you do you know people who are worried about um augmenting this lifespan well what if it had been 20 or 30 you know then what and you say well you know that's too short why is that too short it could have been anything it's but my point is it's completely arbitrary and so and so let's not um base our decisions about what um what ought to be on where the process of random mutation and selection has has dumped us so that's you know I kind of I think that's important to say um be be beyond that I mean we're certainly involved in uh some some uh some research on aging and the role of bi electricity in aging and kind of the uh uh the degradation of bioelectric patterning information with age and all that but I think more importantly is the my my Weather Vein for all of this stuff is plaria so so I think you know plaria basically I think hold hold the answer to all of life's big questions and so uh I mean the the asexual form of um plenaria that we have they do not age they they they there is no there's no evidence of an old planarian they simply

they they can go forever so these ideas  
that it's some sort of inevitable  
accumulation of errors or it's tired or  
it's you know uh clearly there are ways  
around it that's the one thing that's we  
we have an existence proof uh that uh  
and then there's a few other animals  
that do but planaria I think are the  
most interesting uh there there is  
obviously a strategy that gets around it  
and I think it's on us to figure out why  
that is and I think it's absolutely not  
a coincidence that planaria are also  
extremely regenerative in addition to  
being immortal they're extremely  
regenerative they're cancer resistant  
and they have an incredibly noisy genome  
because they don't clean their genome in  
every generation like we do they right  
somatic mutations just if you if it  
doesn't kill the cell it just basically  
expands into the next generation BL can  
be mixed supposed every celllet have a  
different number of chromosomes they're  
just a complete mess and and those are  
the animals that are that are highly  
regenerative cancer resistant and  
immortal isn't that amazing isn't that  
right yeah and and so there so so  
finally after after years of being like  
feeling scandalized about this you know  
the importance of the genome and all  
that and and why is the animal with the  
with the craziest genome also have all  
these amazing capacities um I think we  
actually finally have have a lead on  
on what's going on there but um but I  
think think I think that's that's what

we need to look at in terms of Aging is  
is is I I think if regenerative if the  
if the question of regenerative medicine  
were solved aging would be solved as a  
byproduct excellent yeah I mean it makes  
me think  
um in some ways I I don't know about the  
the function of planaria um you know that  
sort of the worms people can search up  
pictures of planaria um but it makes me  
think that they like the in many ways  
the there just take ourselves for  
example what makes a human a human is in  
some ways the fact that the the  
preference prior that we have or our  
limited functional our functionality is  
limited right like I am not going to be  
at 39 38 37 and 36 degrees Centigrade  
with equal  
probability but it sounds to me like  
planaria able to sort of metamorphosize  
in some way some of their preferences  
because as you say their genome is  
shaping is that is that kind of a  
misconstrue of what's going on is it  
that with or that with the sort of multi  
the the multiplicity of the genome the  
messy genome comes a multitude of  
function or is it that the function is  
still  
constrained yeah I'm I'm going to I'm  
going to give you uh and and and  
unfortunately have to we have to run in  
a couple minutes but I'll just give you  
a very quick uh story of what I think is  
going on in planaria uh I think uh and  
and this is um Steve Steve Frank told me  
an interesting analogy to this

apparently uh you know when when raid arrays became popular right so computer um computer discs that are that makes it make copies of themselves so that you know if something goes wrong he basically said that when raid erase became popular the quality of disk media went down because it wasn't as important anymore to have to have your Hardware be super reliable because you've got this this the system on top of it I think what what's happened in planaria in that lineage is that it has uh really uh come to grips with the fact that the hardware might be junk and that all the evolutionary effort went into cranking the algorithm basically the idea that we we're going to assume from the beginning that you're full of mutations you're unre it's it's unreliable hardware and we're going to assume it's unreliable and so what we're going to do is do our best to have a system that is so highly regenerative that no matter what's going on we can we have St a pretty good chance of making a good planarian the the the development of planaria is called a chaotic mode where it's a total mess and then it sorts itself out into a proper worm um I think uh and and and here's why I think it happened and and we have computational models that show that show how this works imagine that you you have a uh certain competencies to fix defects for example like in the Frog we know that if if the mouth is moved off a little bit in development it'll soon make its way back to where it

needs to be it's sort of fixed fixes  
itself so so in evolution if you have a  
system that has a little bit of  
Competency to fix itself when that  
individual comes up for selection and  
selection sees that okay this is a great  
individual what selection doesn't know  
is is it great because the genome was  
amazing or is it great because the  
genome was actually so so but it fixed  
itself you don't know once you don't  
know you're it becomes really difficult  
to reward for the best genomes what you  
can reward is for the best phenotypic  
fitness for the best outcomes and as  
soon as you do that that makes the  
problem worse because then the  
competency goes up then then it becomes  
even harder to know what your genome was  
so so you so you start off with this  
ratchet with this positive feedback loop  
and I think in *planaria* it went all the  
way and in another species not so far  
and it's some like *celegans* maybe not at  
all but but but that that idea that that  
the the more competency of of um uh kind  
of uh uh reparative regenerative  
developmental process it it it you can  
you can eventually if you take it to its  
logical conclusion you can say fine  
we're going to assume that the that the  
structural genome is going to be all  
kinds of noise and we're going to come  
up with an algorithm that can keep it  
clean keep the the morphology clean  
assuming that that's assuming that  
that's the case and I think *planaria*  
just like cranked that knob all the way

and salamander sort of do it and and you know and maybe neems don't do it much that is very very cool it makes me want to be a biologist because it the question if we had more time and I know we don't is well where on Earth is that regenerative mechanism coming from if it's not encoded in the genome somewhere but we won't go into it it just sounds to me like there are other ways to maximize model evidence to use these active influence words to maximize model evidence or to return to an attractor State Beyond genomic stability and that's really fascinating because I think people think well the the priors the So-Cal phenotypic priors are just what evolution has endowed Us in terms of our DNA yeah and and and and so if we wanted to have another discussion this is where we would talk about xenobots and anthr robots and these things that have never been under selection at the organism level here you know in the history yeah I mean I've also been I'll finish up here I've been reading a little bit of Evan Thompson in mind in life and he talks about how well the the autopoietic coupled system is not going you can't discover that dynamic in the genome because there are things that are also selected over time which which refer to the external Dynamic and which refer to the world in which we are born and thrown into and that's really important as well and I wonder whether what that interplay means in terms of regeneration and and morphogenesis but



as as you're all aware and I'm aware  
we're Limited in time so we should I'd  
absolutely love to do this again I mean  
it was absolutely fascinating um guys  
thank you uh just if you have a minute  
it' be great to know where people can  
find your details and also like what you  
guys have got coming up in the pipeline  
that people can be aware of France you  
want to go first sure I mean you can  
basically look at our names and and you  
know SC and get some of our papers um we  
have a great website the for 11 La there  
has a bunch bunch of our research come  
eliminated um my uh my papers coming up  
we actually look in some of these  
questions in an anual organism in volvox  
that's going to be happening soon I'm  
very excited about that uh  
because all all the works published for  
me right now is a lot of the the  
theoretical simulations and talking  
about the concepts but I'm excited about  
most of my PhD work was trying to apply  
this to an organism and see whether or  
not there's anything there um so that be  
fantastic and Mike yeah so so the lab  
website is at uh dr mike 11.org one word  
dr mike 11.org and so that's where all  
the peer reviewed stuff the the papers  
the protocols um data sets software  
everything is is there and then kind of  
a little bit more out there some  
thoughts are at a Blog called thought  
forms. lifee and that's just my that's  
just my personal one I'll make sure  
everything goes in the uh video  
description so people can check it out

there but guys thank you again thank you  
so much yeah yeah great conversation  
thank you so much thank you it's  
pleasure